# Predicting Urban Taxi Demand and Smart Charging Strategies for Electric Fleets

Advanced Analytics and Applications

Team 1


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# 1 Introduction

## 1.1 Problem Description

Urban taxi demand is highly uneven across both space and time. In a city such as Chicago, trip volumes vary by neighborhood, hour of day, weekday, weather conditions, airport activity, commuting patterns, and local points of interest. For a fleet operator, this creates a practical planning problem: vehicles should be available where passengers are likely to request rides, while unnecessary idle time, empty repositioning, and inefficient charging should be reduced.

This project uses Chicago taxi trips as a proxy for urban ride-hailing demand. The core problem is to understand and predict how many trips will start in different parts of the city during a given time window. In this report, demand is therefore defined as the number of taxi pickups per spatial unit and time interval.

## 1.2 Business Goal

The business goal is to support better operational decisions for a taxi or ride-hailing fleet operator. More accurate knowledge of where and when demand occurs can help operators position vehicles in high-demand areas, reduce passenger waiting times, avoid unnecessary idle time, and improve the utilization of available vehicles. For an electric fleet, this planning task is connected to charging decisions: vehicles need enough energy to serve expected demand, but charging should be scheduled in a cost-efficient way.

From a managerial perspective, the project aims to turn historical trip data into practical recommendations for fleet positioning, demand planning, and charging strategy. The final value of the analysis is not only whether a model predicts demand accurately, but whether the results can inform decisions that improve service reliability and operational efficiency.

## 1.3 Data Mining Goal

The data mining goal is to build a spatio-temporal demand prediction task from raw taxi trip records and external contextual data. Individual trips are aggregated into panels by time window and spatial unit, and the target variable is the number of pickups in each panel cell. Calendar variables, weather data, and spatial identifiers are used as explanatory features.

The project evaluates whether machine learning models can predict aggregate taxi demand at useful resolutions. In particular, Support Vector Machine models and neural networks are trained and compared against simple time-profile baselines. Model performance is assessed using appropriate error metrics and benchmarking, with attention to whether the models perform well not only overall but also in the high-demand areas that matter most for fleet operations. In a second analytical task, a reinforcement learning approach is used to study how charging decisions for an electric taxi can be optimized under operational constraints.

# 2 Data Description

The analysis combines taxi trip records, hourly weather data, and spatial reference data for Chicago. The taxi and weather data cover 2024-01-01 through 2026-05-31. The taxi data cuts off at the end of May 2026 rather than the pull date because the portal has a reporting lag of several weeks — June 2026 held only 31 rows instead of the roughly 600,000 expected, which would otherwise look like a demand crash in the prediction models.

## 2.1 Taxi Trip Data

The main data source is the Chicago Taxi Trips dataset from the City of Chicago Data Portal (dataset `ajtu-isnz`). Each row represents one reported taxi trip by a licensed Chicago taxi. The raw export contains 16,086,180 rows and 21 columns before cleaning.

The dataset contains temporal variables (`trip_start_timestamp`, `trip_end_timestamp`), trip characteristics (`trip_seconds`, `trip_miles`), price components (`fare`, `tips`, `tolls`, `extras`, `trip_total`), categorical fields (`payment_type`, `company`), and anonymized identifiers (`trip_id`, `taxi_id`). Pickup and dropoff locations are reported through census tracts, community areas, and centroid coordinates.

Chicago reports locations at two spatial resolutions. Community areas are coarse city zones; there are 77 official community areas covering Chicago. Census tracts are more fine-grained, but they are frequently missing because the city masks low-volume tracts for privacy. The centroid latitude and longitude columns therefore mix resolutions: where a census tract is available, the centroid belongs to the tract; otherwise, the coordinates fall back to the community-area centroid. This limitation is important for all spatial analyses and is handled explicitly during data preparation.

## 2.2 Weather Data

Hourly weather data for Chicago are obtained from the Open-Meteo archive API and stored in `data/chicago_weather_2024_2026_hourly.csv`. The file contains 21,168 hourly observations and 9 variables, covering the same period as the taxi data. The variables include temperature, apparent temperature, precipitation, snowfall, wind speed, cloud cover, WMO weather code, and a human-readable weather description.

Weather is included because it is an external factor that may influence travel behavior. It is also suitable for demand prediction because weather variables are exogenous and can be known before the prediction period. In later modeling steps, weather observations are joined to taxi trips and demand panels through their timestamps.

## 2.3 Spatial Reference Data

Several spatial datasets are used to interpret and aggregate the taxi locations. The official Chicago community-area boundaries contain 77 polygons and are used for maps, spatial joins, and community-area-level demand aggregation. Census-tract boundaries contain 878 polygons and provide a more detailed spatial layer where tract-level taxi information is available. The city boundary is used as the geographic frame for maps and spatial filtering.

The project also uses point-of-interest data from OpenStreetMap, collected through Overpass Turbo and stored as `data/poi_chicago.geojson`. These data contain locations such as transport facilities, amenities, tourism-related places, and other urban activity points. They are used as contextual spatial information and can help explain why some areas generate more taxi demand than others.

## 2.4 Granularity and Limitations

The raw taxi data are trip-level observations. For prediction, they are aggregated into spatio-temporal panels where the target variable is the number of taxi pickups per spatial unit and time window. The analysis uses different spatial resolutions, especially community areas and census tracts, and different temporal resolutions such as hourly and four-hour windows.

The final cleaned taxi dataset described in the preparation section contains 14,465,650 rows and 27 columns. The most important limitations are the privacy masking of census tracts, the reporting lag at the end of the portal data, and the fact that taxi trips are used as a proxy for broader ride-hailing demand rather than observing ride-hailing platforms directly.

# 3 Brief data preparation details

The raw taxi export contains 16,086,180 rows across 21 columns. The cleaning pipeline concatenates the 2024, 2025, and 2026 CSV exports, applies the steps below, and writes the result out as a single Parquet file. In total, 10.1% of trips (1,620,530 rows) are removed:

1. **Duplicate removal.** Trips are deduplicated on `trip_id` before any other filtering (775 rows removed), keeping the first occurrence. Rows without a `trip_id` are set aside first and rejoined afterward, since `drop_duplicates` would otherwise treat several genuine but ID-less trips as duplicates of each other.
2. **Handling missing spatial identifiers.** Chicago reports location at two resolutions: `census_tract` (fine-grained) and `community_area` (77 coarse zones covering the whole city). Tracts are missing for over half of trips — not randomly, but because the city masks any tract with too few trips, for privacy. Community areas are far more complete (98.2% pickup, 91.6% dropoff), since they're only missing when a trip starts or ends outside city limits. Rather than dropping those out-of-city trips, we keep every row and add `pickup_flag`/`dropoff_flag` columns marking whether an area is present, so later spatial analyses can filter on the flags without losing the trips for everything else.
3. **Resolution-consistent centroids.** The centroid latitude/longitude columns silently mix resolutions: where a tract exists, they hold its centroid; where it doesn't, they fall back to the community area's centroid. We derive a clean area-to-centroid mapping from exactly the rows where the raw coordinate already is the area centroid and join it back as separate `*_ca_centroid_lat/lon` columns, giving a resolution-consistent coordinate for any area-level aggregation.
4. **Removal of incomplete trip records.** Trips missing `fare`, `trip_total`, `trip_seconds`, or `trip_miles` are dropped (34,687 rows), since no meaningful analysis is possible without them.
5. **Outlier and plausibility filtering.** Negative `tips`/`tolls`/`extras` would be removed, though none occur in practice — zero is legitimate, for example for cash trips that report no tip. Trips below Chicago's legal minimum fare ($3.25 flag-drop) or shorter than 60 seconds are treated as test or aborted entries and removed, together with trips above the 99.9th percentile of distance (43.0 miles) or duration (165 minutes). These bounds account for the bulk of all removed rows (1,585,068 of 1,620,530).

`payment_type` and `company` are left untouched throughout; their missing values remain as `NA` and are handled per analysis.

The cleaned dataset has 14,465,650 rows and 27 columns, spanning 2024-01-01 to 2026-05-31.

# 4 Demand Prediction

## 4.1 Problem Description

We predict taxi pickup demand, defined as the number of trips originating in a spatial unit within a fixed time window. Two model families are used: Support Vector Regression (SVR) and

feedforward neural networks (NN). Both are evaluated across several spatio-temporal resolutions: community area at 4-hour and 1-hour windows, and census tract at 4-hour windows. Results below are organized by resolution so the two model families can be compared directly wherever both are available; a synthesis of findings across resolutions and models follows in Section 4.5.

## 4.2 Modeling Approach

Both model families share the same task framing: a single **pooled model per experiment**, trained across all spatial units at once, with spatial identity encoded as one-hot fixed-effect dummies rather than fit as separate per-unit models. This scales to hundreds of census tracts without multiplying training time, and lets panel-construction and evaluation code be reused across resolutions. Two families were tried for complementary reasons: SVR (linear and kernelized) is cheap to fit and, in its linear form, interpretable via coefficients; a feedforward network can in principle learn non-linear interactions between spatial identity and time features that a linear model cannot without hand-built interaction terms, at the cost of a larger tuning surface and no direct coefficient interpretation.

The central obstacle every model in this section has to overcome is that demand is not one distribution, it is dozens (or hundreds) of very different ones. Per-unit demand varies roughly three orders of magnitude across community areas, and the same imbalance only gets sharper at the finer-grained census-tract level, where far more units compete for the same total demand. Each unit also has its own daily and weekly rhythm, not just its own scale: an airport's demand follows flight schedules and runs around the clock, while a downtown office district peaks at commute hours and goes quiet overnight. A pooled model has to represent all of that with one set of parameters.

This is a genuinely hard bind for SVR specifically. `LinearSVR` fits a single global coefficient vector, so it can only find one compromise direction that trades off across every spatial unit at once; it has no mechanism to give a high-demand downtown area its own scale and its own rush-hour shape independent of a quiet residential area's. A kernel SVR (RBF, polynomial, sigmoid) could in principle represent that kind of heterogeneity through non-linear interactions between spatial identity and time features, but its training cost scales quadratically to cubically in the number of rows, which puts the full multi-hundred-thousand-row pooled training set out of reach. In practice this forces kernel variants onto a small subsample (10,000 rows), nowhere near enough to represent dozens of areas' worth of individual dynamics, and, as the results below show, does not solve the underlying problem even where the tuning itself is trustworthy.

Both use the same exogenous feature set: weather (temperature, precipitation, snowfall, wind, cloud cover), cyclical time encodings (sine/cosine pairs for hour-of-day and day-of-week, two harmonics each to capture the bimodal morning/evening rush), month, and weekend/holiday indicators. Per the constraints of this course, no lagged or autoregressive demand features are used. Only inputs known in advance of a prediction are included. Both regress on `log1p(count)`, with predictions back-transformed via a clipped `expm1`, since raw demand is heavily right-skewed (Figure 6; median per-unit demand spans roughly three orders of magnitude); without this transform, squared-error loss would be dominated by a handful of high-demand windows.

At census-tract resolution, one-hot spatial dummies for all 878 tracts would be impractical, so tract-level SVR instead reuses the 76 community-area dummies as a coarse spatial layer and adds tract-level covariates as a fine layer: POI count, tract area, and distance to six empirically-derived demand hot spots (identified separately via a Gaussian Mixture Model over pickup coordinates).

## 4.3 Validation Strategy

Both families use a chronological holdout, training on 2024-01-01 through 2025-12-31 and testing on 2026-01-01 through 2026-05-31, over a complete, zero-filled grid of every spatial unit crossed with every time window. This means evaluation includes genuine zero-demand periods, not only observed trips. SVR hyperparameter search uses `KFold` (3, shuffled): since no lagged features are used, there is no autoregressive leakage risk a chronological split would guard against, and shuffled folds give an even seasonal mix. The NN training procedure splits the training period randomly 80/20 into a fit set and an early-stopping set for the same reason.

A **profile baseline** serves as the reference every model is judged against: historical demand per (hour, weekday, month), computed separately per spatial unit. Because it is already computed separately per unit, it sidesteps the heterogeneity problem described in Modeling Approach entirely: it never has to compromise across areas the way a pooled model does, which is the main reason it turns out to be such a persistently hard target to beat throughout this section, not any particular weakness of SVR or the NN as an algorithm.

Two metrics are reported, and they are not computed the same way, which matters for reading the numbers below correctly. The **skill score**, $1 - \mathrm{MAE}_{\mathrm{model}}/\mathrm{MAE}_{\mathrm{baseline}}$ per unit, is a ratio, so each unit's own demand scale cancels out, making it comparable across units *and* across resolutions. It is aggregated two ways: **demand-weighted skill** (the headline number throughout Results) is the weighted average of every unit's own skill score, weighted by that unit's *median* demand, so the highest-volume units drive the number, matching a fleet-allocation use case where getting the busiest areas right matters most. The **unweighted median** (every unit counted equally) is a guard rail against a model that only wins on a few large units and loses everywhere else. Skill score is currently computed for SVR only. The **weighted RMSE** is a separate, absolute, trips-per-window number reported by both families: pooled RMSE with each row weighted by its unit's train-period *mean* demand, not the median used for demand-weighted skill above. It answers a different question, how large the errors are in absolute trips once high-volume rows are up-weighted, rather than skill's relative one, how much better than the baseline. It is therefore the metric used for direct SVR-vs-NN comparison below. Plain, uniformly-weighted pooled $R^2$/MAE/RMSE, and the median per-unit $R^2$ that recurs throughout Results, are reported only as supporting context, not headline numbers: demand varies roughly 1000-fold across units, so unweighted pooled numbers are dominated by the busiest few, and per-unit $R^2$ is noisy for the many near-zero-demand units where there is little variance for any model to explain in the first place.

## 4.4 Results

### 4.4.1 Community Area, 4-Hour Resolution

**SVR.** The profile baseline is a much harder reference per-area than pooled numbers suggest (Figure 7): pooled $R^2$ is 0.933 (weighted RMSE 139.02), but median per-area $R^2$ is only 0.208, with 21 of 77 areas negative, mostly low-demand areas where near-zero counts leave little variance for any model to explain. A pooled $R^2$ is dominated by the handful of highest-volume areas, the heterogeneity problem from Modeling Approach showing up directly in a single number, which is exactly why skill score, not pooled $R^2$, drives the comparisons below.

A pooled `LinearSVR`'s demand-weighted skill is **-0.827** (weighted RMSE 254.94, nearly double the baseline's): it systematically underperforms the baseline specifically in the highest-demand areas, despite winning on a majority of areas by unweighted median skill (-0.017), since every area contributes an equal number of training rows regardless of demand, so the shared coefficients fit mostly to the low-demand majority (Figure 1). Refitting with sample weights

proportional to $\sqrt{\text{median demand} + 1}$ raises demand-weighted skill to -0.675 (weighted RMSE 231.91) but pushes 53 of 77 areas negative (up from 40): reweighting toward the areas the linear model handles worst comes directly at the expense of the areas it handled comparatively well, a trade-off, not a fix, because one global coefficient vector still cannot serve both regimes at once.

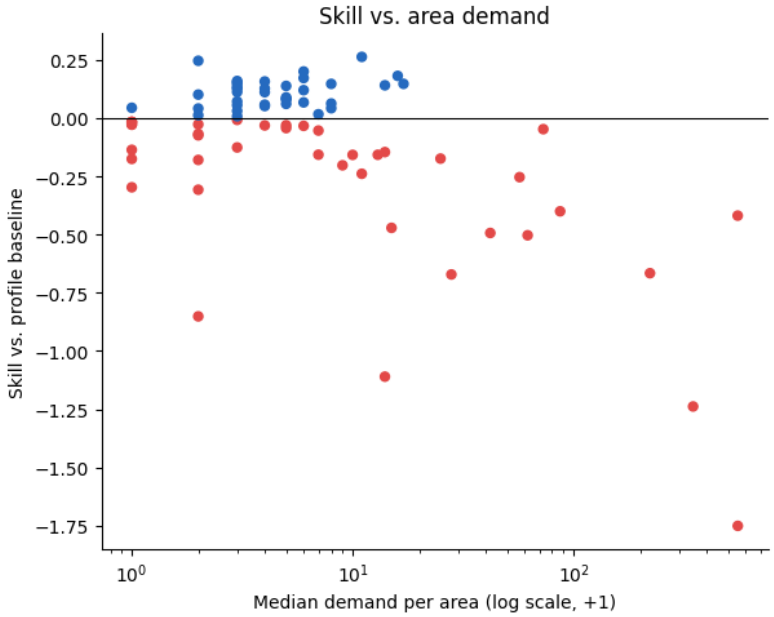


Figure 1: Per-area skill vs. median demand (log scale). Skill trends negative as demand increases: the model loses specifically in the areas that matter most for fleet allocation.

Feature-coefficient analysis shows what the pooled model actually leans on (Figure 2): the dominant coefficient by a wide margin is the primary daily-cycle harmonic (`hour_cos`), several times larger than any other feature, followed by its second harmonic and the weekend indicator. Every weather variable (temperature, wind speed, cloud cover, precipitation, snowfall) sits at the bottom of the ranking with a coefficient close to zero: the model has correctly identified that demand is driven by the clock, not the weather, even though it still loses to the baseline on the highest-demand areas (a contrast with the census-tract finding below, where a strong static predictor takes over instead and the time signal disappears).

Kernel SVR (RBF, polynomial, sigmoid) was compared on the same 10,000-row subsample necessity established above. A first pass caught a methodological trap: scaling the community-area dummy columns flipped which kernel looked best, since RBF's distance-based kernel was dominated by the inflated dummy scale, so dummies were left unscaled from then on, as with `LinearSVR`. The subsample winner (polynomial, degree 2) and RBF were then tuned with `GridSearchCV` (`KFold`, 3, shuffled, per Validation Strategy) over a small grid (`C`/`degree` for polynomial, `C`/`gamma` for RBF). Tuning here is trustworthy, not unstable: cross-validated scores track the test set closely for both (polynomial: CV $R^2 = 0.865$, test $R^2 = 0.896$; RBF: CV $R^2 = 0.891$, test $R^2 = 0.892$). It still does not solve the underlying problem, though: per-area demand-weighted skill stays negative for both (polynomial -0.221, 49 of 77 areas negative; RBF -0.232, 46 of 77 negative), the same pooled-$R^2$-looks-good, per-area-skill-does-not gap as plain `LinearSVR` above. Kernel tuning is therefore documentation only, not the SVM entry carried forward.

Fully separate per-area models test whether pooling itself is the problem: one dedicated model per area, fit only on that area's own rows, with no coefficients shared across areas of different scale. This only makes sense above a demand threshold (median $\geq 10$ trips/window, 20 of 77 areas, Figure 11): below it, an area's target has too little variance for any regression model

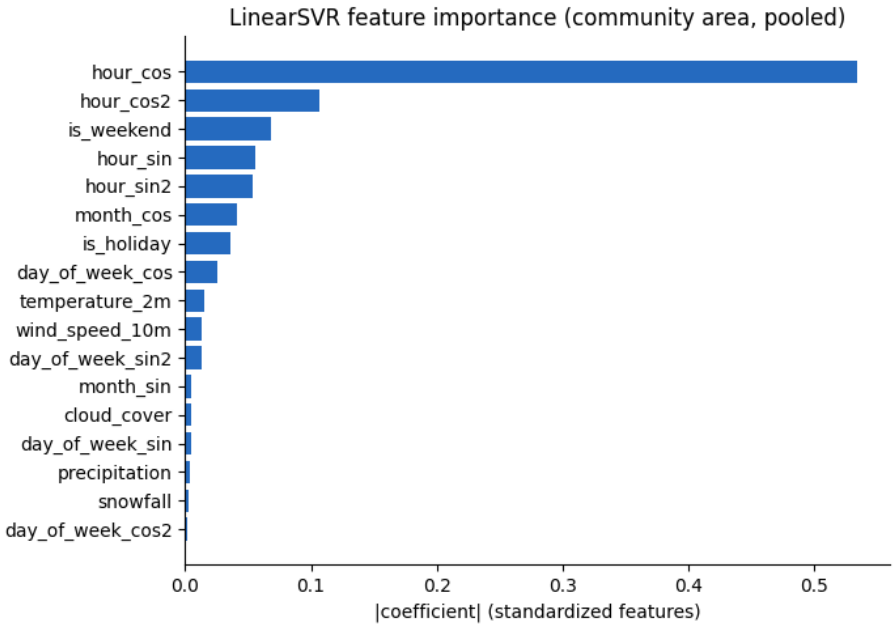


Figure 2: LinearSVR feature importance, community area. The primary daily-cycle harmonic (`hour_cos`) dominates by a wide margin; every weather variable sits at the bottom of the ranking.

to meaningfully beat its own mean, so the 57 excluded areas keep using the pooled and baseline predictions computed elsewhere instead. This costs real computation: hyperparameters are tuned separately per area (`GridSearchCV`, `TimeSeriesSplit`, 3 folds), so tuning becomes 20 grid searches rather than one, roughly 20 to 45 minutes for the RBF grid alone. In exchange, exact kernel SVR, infeasible on the full 337,722-row pooled set, becomes entirely tractable per area, since each area's own history is only around 4,400 rows. The comparison to pooling is the point, and it is a clear, if narrow, win. Untuned per-area `LinearSVR` alone reaches demand-weighted skill -0.112, already well ahead of the pooled model's -0.827, before any tuning. Tuned per-area RBF SVR then becomes the single configuration in the entire study, pooled or per-area, with positive demand-weighted skill: **+0.012** (Figure 12); it is no coincidence that the one configuration that wins also abandons pooling. The win is thin, though: median skill is -0.002 and 10 of the 20 areas remain individually negative, even with pooling removed.

Table 1 collects all ten SVR configurations tried at this resolution side by side. The pattern is consistent across every row but one: pooled or per-area, tuned or not, almost everything loses to the baseline on demand-weighted skill, and the single exception wins by a thin margin.

Table 1: Full model comparison, community area, 4-hour resolution. Per-area models cover only the 20 areas above the demand threshold, not all 77.

| Model | Units | Dem.-weighted skill | Median skill | Weighted RMSE | Pooled $R^2$ |
|---|---|---|---|---|---|
| Profile baseline | 77 | 0.000 | 0.000 | 139.02 | 0.933 |
| LinearSVR, pooled | 77 | -0.827 | -0.017 | 254.94 | 0.778 |
| LinearSVR, weighted | 77 | -0.675 | -0.052 | 231.91 | 0.814 |
| LinearSVR, tiered | 77 | -0.732 | 0.000 | 243.60 | 0.800 |
| RBF, tuned | 77 | -0.232 | -0.027 | 176.93 | 0.892 |
| Polynomial, tuned | 77 | -0.221 | -0.049 | 171.71 | 0.896 |
| LinearSVR, per-area | 20 | -0.112 | -0.091 | 156.93 | 0.905 |

| Model | Units | Dem.-weighted skill | Median skill | Weighted RMSE | Pooled $R^2$ |
|---|---|---|---|---|---|
| LinearSVR, per-area, tuned | 20 | -0.125 | -0.116 | 161.10 | 0.901 |
| RBF, per-area | 20 | -0.135 | -0.105 | 163.37 | 0.897 |
| **RBF, per-area, tuned** | 20 | **+0.012** | -0.002 | 143.80 | 0.921 |

**NN.** *To be added.*

**Shared finding.** *To be added once NN results are finalized.*

### 4.4.2 Community Area, 1-Hour Resolution

**SVR.** This resolution is a deliberate scaling stress test: same 77 areas, roughly 4x the rows, heavier zero-inflation, and a harder baseline too, since it now resolves 24 hourly buckets instead of 6 (baseline weighted RMSE 39.46). A pooled `LinearSVR`'s demand-weighted skill is **-1.016** (weighted RMSE 76.96), a worse deficit than the 4h resolution's -0.827 for the same model type: finer time resolution sharpens the heterogeneity problem rather than easing it, since the same 77 areas must now be told apart across four times as many hourly buckets.

Kernel comparison and tuning on a 10,000-row subsample favor RBF: after tuning, RBF reaches demand-weighted skill -0.331 (weighted RMSE 54.53), polynomial reaches skill -0.361 (weighted RMSE 56.14): the best 1h configurations, though both still negative.

Demand-tiered and per-area models were attempted but dropped at this resolution: per-area `GridSearchCV`'s parallel fan-out, on top of the larger 1h panel, exceeded available memory.

### 4.4.3 Census Tract Resolution

Only 44.6% of trips carry a true tract-level location; the rest are masked to their community area's centroid for privacy whenever a trip's originating tract had too few trips in that period. Masking is far from uniform: retention ranges from 0% to 70.9% by community area, with a clean gap in the distribution between roughly 5% and 7%. Areas below that gap have too little tract-precision data to support any tract-level model at all. This is not a modeling choice but a data-availability ceiling, leaving **179 of 878 tracts, across 11 community areas**, in scope (Figure 13); the remaining areas fall back to the community-area-resolution model. Within scope, counts are built from tract-precision trips only (never the masked fallback points, which would otherwise spike demand artificially onto whichever tract contains a community area's centroid) and scaled by each area's inverse retention rate to estimate total demand. Given the smaller per-unit training data at this resolution, only a pooled `LinearSVR` is fit; kernel SVR is not attempted, following the same compute-cost reasoning established at community-area resolution, now at an even smaller per-unit scale. No NN model has been attempted at this resolution. A census-tract, 1-hour combination was deliberately not attempted either: it would stack both axes that make pooling hard at once, 179 units instead of 77 and the four-fold finer time buckets that already sharpened the problem at community-area resolution (Section 4.5), on top of tract-level data that is already thinner per unit than the community-area panel.

**SVR.** The profile baseline is strong here too (pooled $R^2 = 0.860$), though the per-tract split is noisier than community-area's (Figure 14): median per-tract $R^2$ is only 0.170, with 38 of 179 tracts negative, the same pooled-versus-per-unit gap seen at community-area resolution, now wider because there are more than twice as many units to average over.

Pooled `LinearSVR` fares far worse than at coarser resolutions. Demand-weighted skill, the headline metric everywhere else in this report, is not available here: the weighting computation returns `NaN` for this model, a data issue not yet root-caused (see Limitations), so unweighted median skill is reported in its place: **-0.259**, with 90 of 179 tracts negative. Weighted RMSE is 355.19, more than double the baseline's 133, and pooled $R^2$ collapses to 0.026. This is the clearest illustration in the whole study of what a single global coefficient vector does when it cannot represent per-unit heterogeneity: rather than fail gracefully, it gives up on time altogether and settles for representing which tract a row belongs to.

Feature-coefficient analysis confirms this directly (Figure 3; Figure 15): GMM hotspot-distance features, which take one fixed value per tract, carry 99.6% of total |coefficient| mass despite being only 6 of the 25 continuous features, while time/calendar and weather coefficients are effectively zero. The model has learned each tract's average demand level from its distance to the nearest hot spot rather than any time-varying signal.

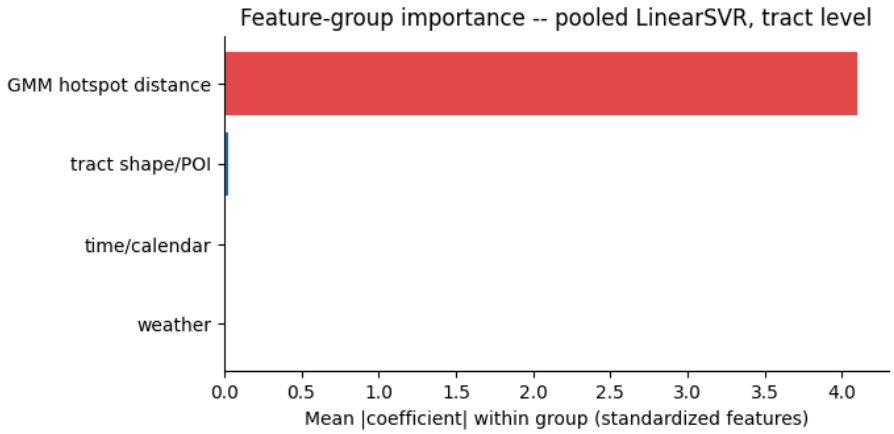


Figure 3: Mean |coefficient| by feature group. Hotspot distance versus time/calendar and weather is the clearest single illustration of the pooling-collapses-to-a-per-unit-constant finding described above.

Actual-vs-predicted plots for the top-3 tracts confirm this directly (Figure 4; Figure 5): predicted demand sits flat near zero across an entire evaluation week while actual and baseline demand both trace the expected daily cycle, a sharp contrast with the community-area case, where the models at least track the shape of demand even while missing the peaks (Figure 10). With 179 pooled units instead of 77, between-unit demand variance dominates the loss even more than at community-area resolution, and the tract-level hotspot-distance covariates give the optimizer an easy per-unit shortcut, at the cost of the temporal signal the model exists to capture.

## 4.5 Cross-Cutting Discussion

Across every SVR variant tried at community-area resolution (pooled, demand-weighted, demand-tiered, per-area), the profile baseline is a surprisingly strong reference specifically where demand is largest (Figure 9; Figure 17), the same heterogeneity problem from Modeling Approach playing out consistently across every variant. It is compounded by an interaction between the `log1p`/`expm1` transform and a squared-error-adjacent loss, which amplifies error multiplicatively: a small log-space miss becomes a large absolute error exactly for the highest counts, a penalty the baseline never incurs since it predicts directly on the raw scale.

**NN.** *To be added.*

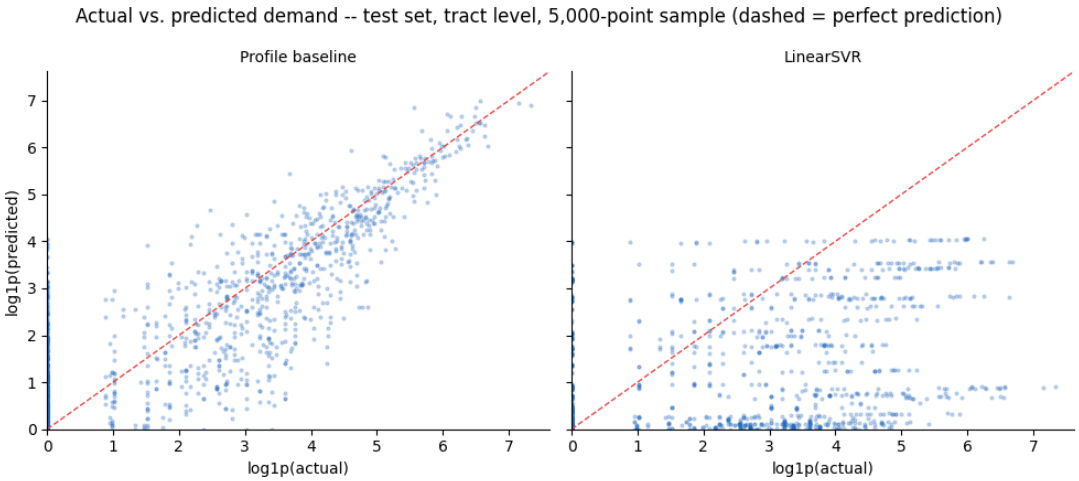


Figure 4: Actual vs. predicted demand, test set, baseline and pooled LinearSVR (log1p scale). LinearSVR predictions cluster into discrete horizontal bands rather than following the diagonal, a signature of a model outputting a level per tract rather than a prediction.

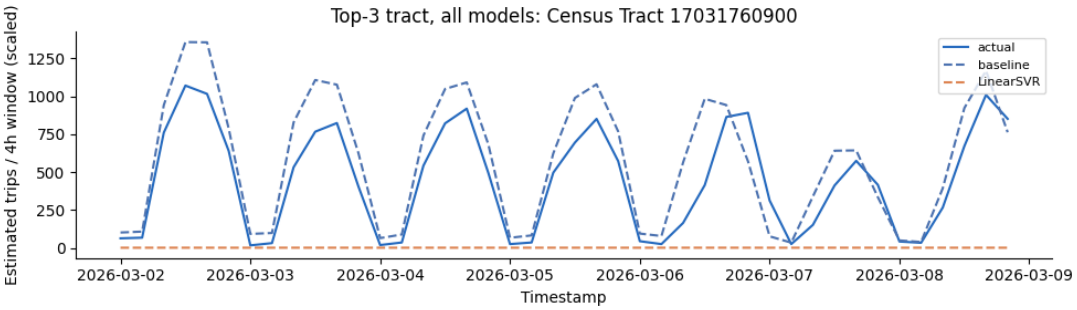


Figure 5: Actual vs. all-models predictions for Census Tract 17031760900 (top-3 by demand), one evaluation week. Predicted demand from LinearSVR sits flat near zero across the entire week while actual and baseline demand both trace the expected daily cycle.

Comparing SVR across resolutions, demand-weighted skill for matching model types is worse at 1h than at 4h (-1.016 vs. -0.827 for pooled `LinearSVR`; -0.331 vs. an untested comparison point for tuned RBF), suggesting finer time resolution makes the high-demand gap harder to close, not easier. This is plausibly due to thinner per-window signal and heavier zero-inflation.

### 4.5.1 Practical Implications for Fleet Allocation

The heterogeneity problem also points toward a more realistic deployment strategy than the one tested here: a business does not need one model that works for every spatial unit, including the many low-demand areas that contribute little to a fleet-allocation decision anyway. It needs good predictions specifically where trip volume is high, since that is where allocation decisions have the most impact.

The one result in this entire study with positive demand-weighted skill, the tuned per-area RBF SVR on the 20 highest-demand community areas (**+0.012**), is also the only one that abandons pooling in favor of a dedicated model per area, for exactly the reasons described above. The practical recommendation this suggests is to not chase one pooled model across all 77 (or 179, or 878) units, but to identify the small set of highest-demand areas that account for most of the fleet-allocation value, fit each one its own model, and leave the long tail of low-demand areas to the simple profile baseline, which already handles them about as well as anything tested here.

## 4.6 Limitations & Outlook

- **SVR and NN are not yet evaluated on matched scope**: differing demand thresholds and baseline definitions (mean vs. median) mean cross-family numbers above are directional, not exact, until reconciled.
- **Skill score is SVR-only**; weighted RMSE is the only metric currently common to both families.
- **Kernel-SVR tuning at community-area resolution is trustworthy but still loses per area**: `GridSearchCV` cross-validated scores track the test set closely (an earlier draft of this section claimed the opposite, based on a since-corrected notebook run), but pooled $R^2$ looking strong (0.89-0.90) still does not translate into positive per-area demand-weighted skill, the same gap seen throughout this section.
- **Demand-weighted skill is unavailable (`NaN`) for the pooled census-tract `LinearSVR`**: the weighting computation fails for this model specifically, not yet root-caused, so that paragraph reports unweighted median skill instead, breaking the otherwise-consistent headline metric used across every other model in this report.
- **Pooled SVR at census-tract resolution underperforms its baseline** (median skill -0.259, pooled $R^2 = 0.026$ vs. baseline 0.860): GMM hotspot-distance features absorb 99.6% of coefficient mass, so the model effectively predicts a constant level per tract and ignores time entirely. This is worth revisiting with per-unit time interactions rather than treated as a data-availability issue.
- **No rolling-origin backtest** for either family: a single chronological holdout is used throughout.
- **1-hour resolution numbers are unverified**: the `03.02` notebook has no saved execution output and no corresponding row in `results/model_results_summary.csv`, so the figures cited in that subsection could not be cross-checked against a live run the way every other number in this section was.

See Section 6 for detailed tables and figures.

## 4.7 Intelligent Charging with Reinforcement Learning

### 4.7.1 Problem Setup

The final part of the project studies a smart-charging problem for an electric taxi. The taxi returns to its home charger at 14:00 and leaves again at 16:00, which creates a two-hour charging window. The agent makes one charging decision every 15 minutes, resulting in eight decisions per episode. At departure, the vehicle must have enough energy for the next shift. If the final state of charge is below the required energy demand, the episode receives a large shortage penalty.

The task is relevant for an electric taxi fleet because charging decisions are not only a technical battery-management problem, but also an operational cost problem. Charging too slowly can leave the vehicle undercharged, while charging too aggressively can create necessary cost. A useful policy therefore needs to balance reliability and cost.

### 4.7.2 Markov Decision Process

We formulate the charging task as a finite-horizon Markov Decision Process. Each episode represents one daily charging window from 14:00 to 16:00.

The state includes the current state of charge, the current time step, the required departure energy, and the battery capacity. In the price-aware extension, the current electricity price and the mean remaining price are added to the state. The action space is discrete and consists of four charging levels: no charging, low charging, medium charging, and high charging. In the implemented notebook, these are represented as 0 kW, 5 kW, 11 kW, and 22 kW.

The transition function updates the state of charge according to the selected charging power and the 15-minute interval length:

$$SOC_{t+1} = \min(C, SOC_t + p_t \cdot 0.25)$$

where $C$ is the battery capacity and $p_t$ is the selected charging power in kW. The reward is the negative charging cost in each step, plus a large terminal penalty if the final state of charge is below the sampled energy demand. This structure encourages the agent to charge enough to satisfy the departure requirement while avoiding unnecessary charging cost.

### 4.7.3 Simulation Environment

The environment is implemented using `gymnasium`, which provides a standard interface for reinforcement learning. At the beginning of each episode, the battery capacity, initial state of charge, and required departure energy are sampled. The energy demand follows a normal distribution with mean 30 kWh and standard deviation 5 kWh, clipped to feasible values.

Two versions of the environment are evaluated. The first version uses a simplified cost function and serves as a controlled baseline setting. The second version extends the problem with real electricity prices from EPEX day-ahead data. In this price-aware setting, charging costs depend on the actual amount of energy charged and the electricity price in the corresponding time step. This extension makes the task more realistic because it allows the agent to shift charging toward cheaper intervals when enough time remains.

### 4.7.4 Learning Algorithm and Baselines

The agent is trained with a Deep Q-Network (DQN). DQN approximates the action-value function $Q(s, a)$ with a neural network and learns from repeated interaction with the simulated environment. The notebook trains the agent over 200,000 time steps and evaluates the learned policy on 1,000 paired test episodes.

The main comparison baseline is a greedy charging heuristic. This heuristic ignores prices and computes how much energy is still missing relative to the departure requirement. It then selects the smallest charging power that can cover the remaining demand within the remaining time. This rule is simple, reliable, and operationally intuitive. For a final version of the analysis, this baseline should be complemented by a true flat-rate strategy, such as charging at a fixed power level until the target energy is reached.

For a subset of episodes, a brute-force optimum is also computed. Since there are only four actions and eight time steps, all $4^8 = 65,536$ possible charging plans can be evaluated for a given episode. This provides a useful benchmark for judging whether the learned policy is close to the best possible charging plan.

### 4.7.5 Results

In the simplified cost setting, the greedy heuristic performs very strongly. It reaches a 100% success rate and is equal to the brute-force optimum on the evaluated subset. The DQN reaches a lower success rate of 96% and a substantially worse mean reward. This indicates that, under the simplified cost structure, reinforcement learning does not add value over the simple rule-based policy. The problem is structurally easy: if prices are almost constant, the best strategy is simply to charge enough to meet the known energy requirement.

In the price-aware setting, the DQN learns to react to electricity prices. Among episodes in which both DQN and greedy succeed, the DQN achieves lower average charging costs than the price-blind greedy heuristic. The notebook reports an average cost reduction of 0.0388 EUR per successful day, or approximately 4.7%. However, this cost saving comes with a lower success rate: the DQN succeeds in 95.7% of episodes, while the greedy heuristic succeeds in 100%. Because shortage is operationally costly, the lower reliability is an important limitation. When the terminal shortage penalty is included in the reward, the greedy heuristic remains the more robust policy.

The policy visualizations suggest that the price-aware DQN tends to charge more aggressively when the current price is low or when the state of charge is below the required demand. When prices are high and sufficient time remains, the agent is more likely to wait. This behavior is consistent with the intended smart-charging logic, but the reliability gap shows that the learned policy still requires further tuning before it can be recommended for operational use.

### 4.7.6 Operational Implications

The results suggest that reinforcement learning is most useful when the charging environment contains meaningful price variation. Under flat or nearly flat tariffs, a simple rule-based strategy is sufficient and easier to explain. Under dynamic tariffs, a price-aware policy can shift charging to cheaper periods and reduce energy cost, but only if it preserves a very high success rate.

For a fleet operator, the key trade-off is therefore cost versus reliability. A policy that saves a few cents per day but occasionally leaves the taxi undercharged is not acceptable

without additional safeguards. A practical implementation should either use a stronger shortage penalty, enforce a hard minimum state-of-charge constraint, or combine the learned policy with a rule-based safety layer.

### 4.7.7 Limitations and Next Steps

The current notebook provides a solid proof of concept, but several improvements are needed for the final analysis. First, the cost function should be aligned with the assignment's quadratic charging-cost formulation, or the deviation should be clearly justified as an extension. Second, the evaluation should include a true flat-rate baseline in addition to the greedy heuristic. Third, the sensitivity analysis should explicitly vary the expected energy demand, for example by comparing scenarios with 25 kWh, 30 kWh, and 35 kWh mean demand. Finally, the price-aware environment should be made fully reproducible by loading stored price windows locally instead of relying on a live API request during execution.

# 5 Conclusion

# 6 Appendix: Detailed Tables and Figures

Figures below are grouped by resolution, in the same order as the Results section, and cover what is not already shown inline there. The most central figures for each resolution (feature importance, the skill-vs-demand relationship, and the flat-prediction evidence at tract level) appear directly in Results instead of here. NN figures are not yet included, since the NN write-up in Section 4.5 is still pending.

## 6.1 Community Area, 4-Hour Resolution

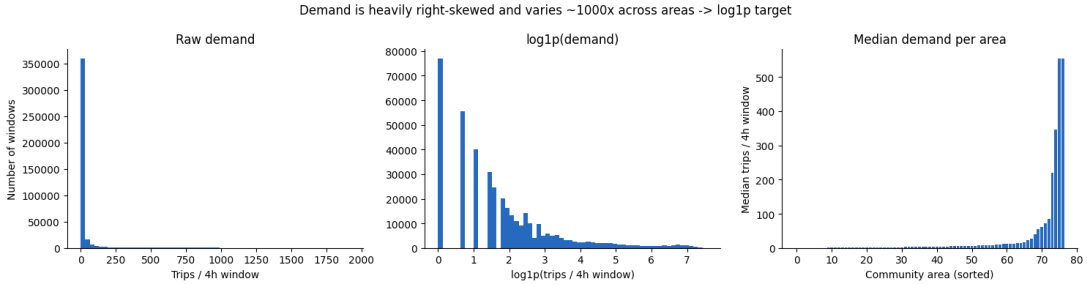


Figure 6: Raw per-window demand is heavily right-skewed, which is why both SVR and NN regress on `log1p(count)` rather than raw counts.

## 6.2 Census Tract Resolution

## 6.3 Pending

- **NN permutation-importance charts** (both time encodings) and NN metric-comparison figures, once the NN sections in Results and Section 4.5 are written.
- **Full model-comparison table**: skill, weighted RMSE, pooled $R^2$/MAE/RMSE per model, per resolution, both families (source data already available in `results/model_results_summary.csv`).
- **1-hour resolution figures**: the `03.02` notebook has not been run to completion, so no saved plot output exists yet to extract from.

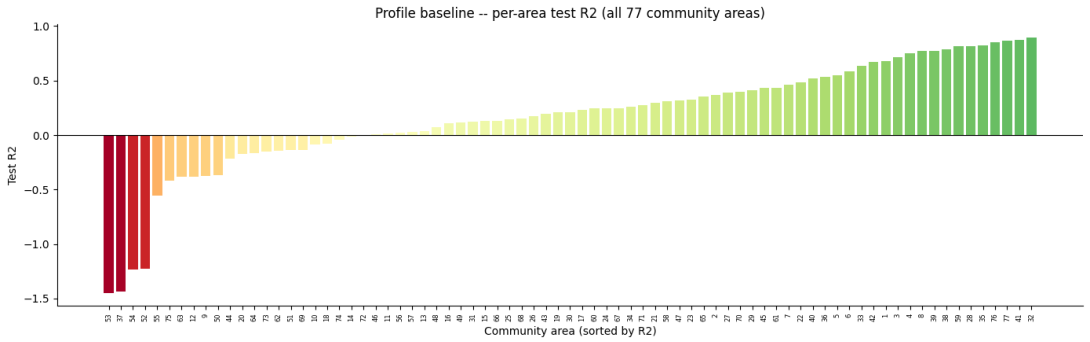


Figure 7: Profile baseline test $R^2$ per community area. Pooled $R^2$ (0.933) is dominated by the highest-volume areas; the per-area distribution is much wider, with several areas negative.

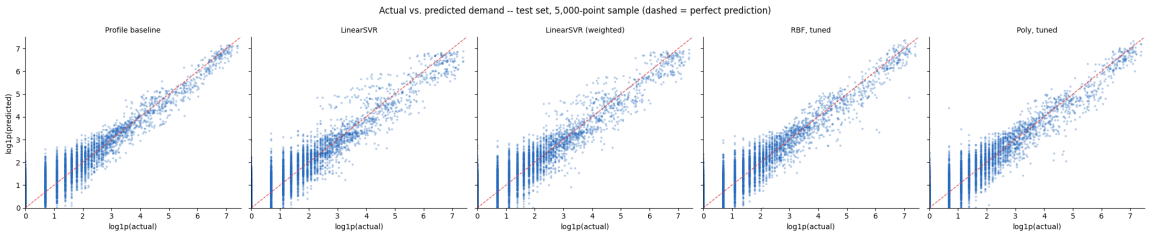


Figure 8: Actual vs. predicted demand, test set, all five community-area models (baseline, pooled LinearSVR, demand-weighted LinearSVR, tuned RBF, tuned polynomial), log1p scale.

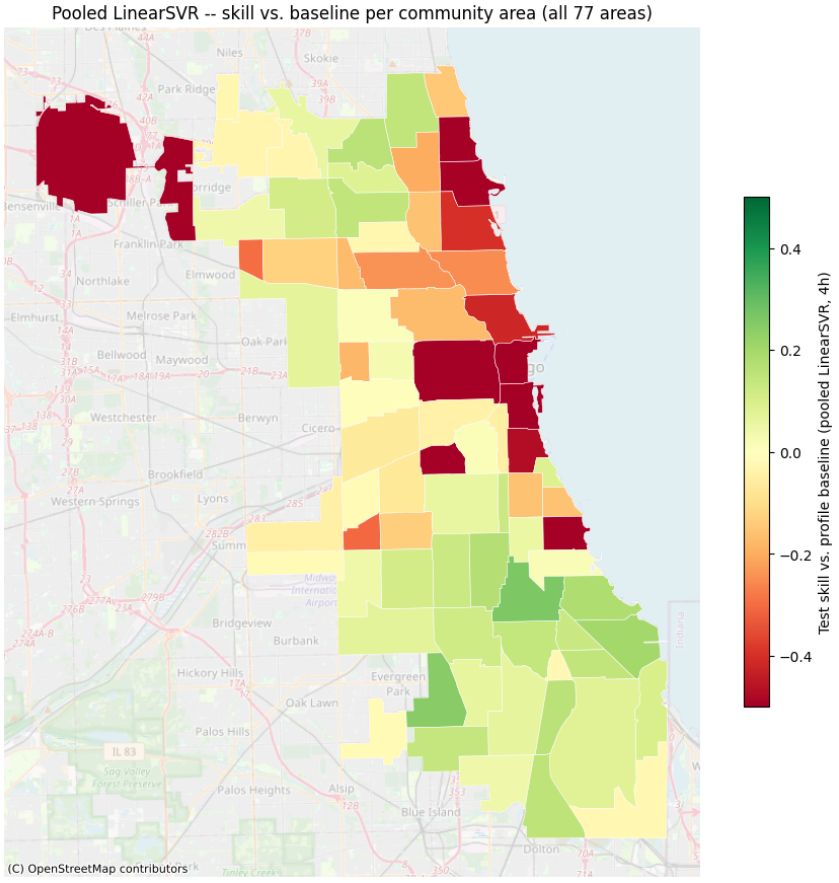


Figure 9: Per-area skill map, pooled LinearSVR, all 77 community areas.

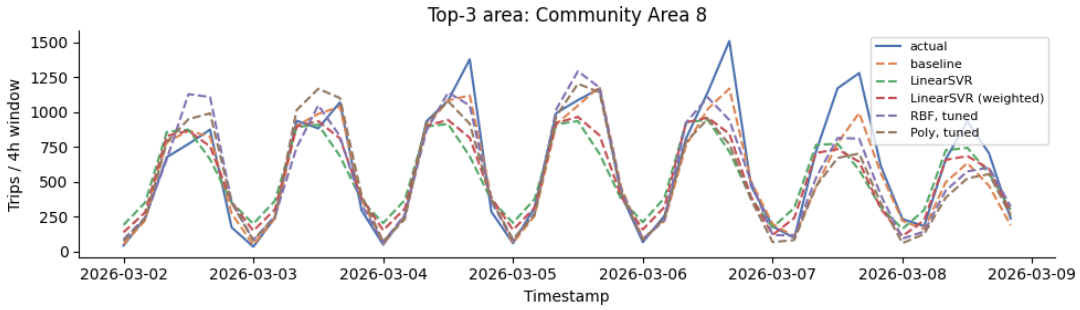


Figure 10: Actual vs. all-models predictions for Community Area 8 (top-3 by demand), one evaluation week. Models track the daily cycle here, unlike the census-tract case below.

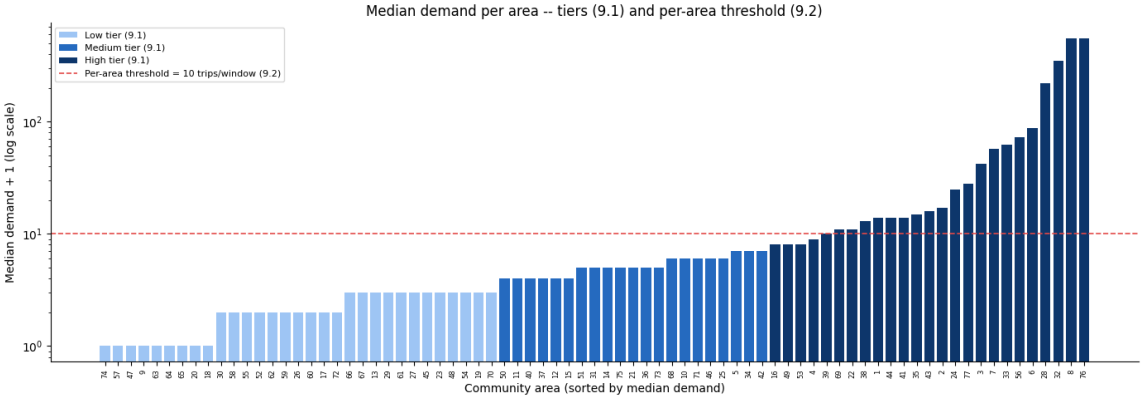


Figure 11: Median demand per area, with the low/medium/high tiers (9.1) and the 10 trips/window per-area threshold (9.2) used to select the 20 highest-demand areas.

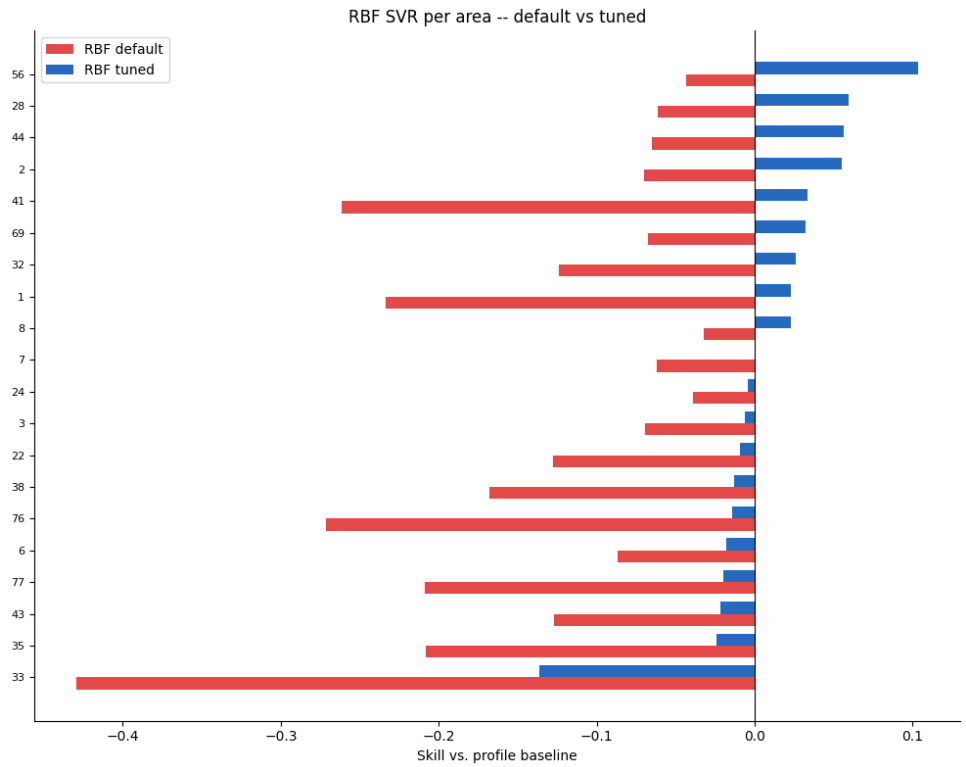


Figure 12: Tuned per-area RBF SVR vs. its untuned default, skill vs. baseline. This is the only configuration in the study with positive demand-weighted skill, and only for the highest-demand areas among the 20.

## 7 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like "R for Data Science" (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

## 8 Conclusion

The conclusion should summarize your findings and suggest future work.

## References

Team, Q. (2024). Quarto: An open-source scientific and technical publishing system. *Journal of Modern Data Science*. https://quarto.org

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science*. O'Reilly Media.

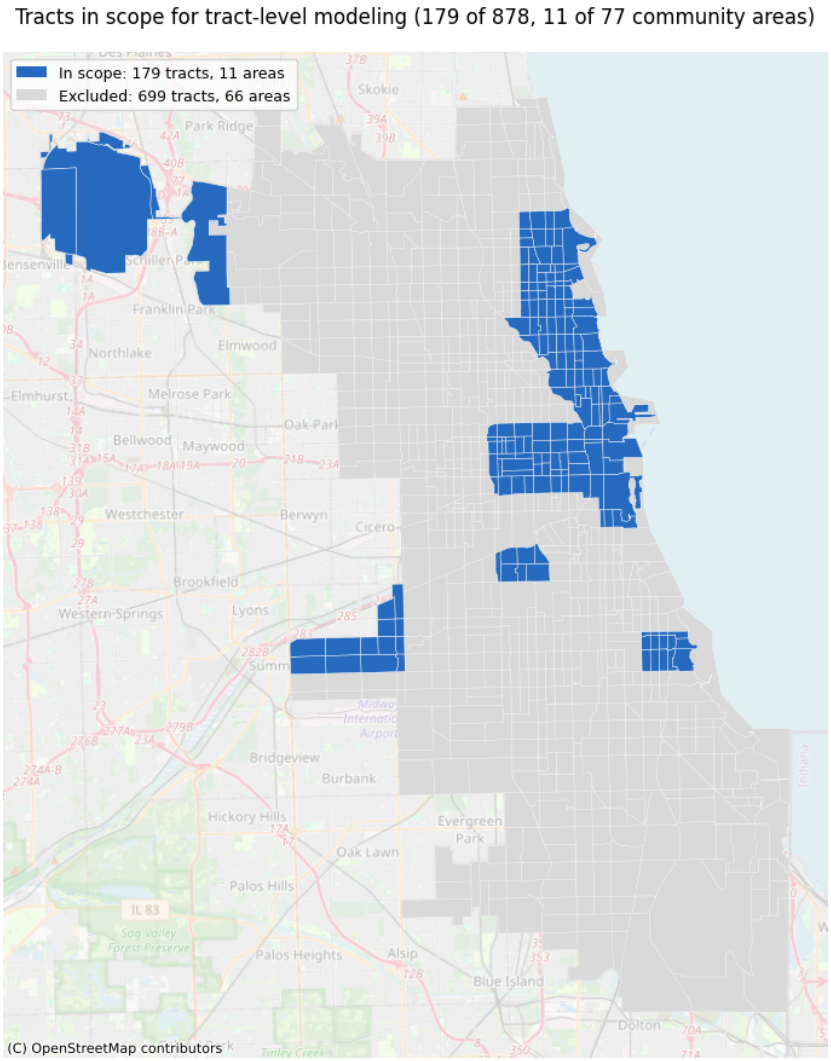


Figure 13: Scope map: the 179 of 878 census tracts with enough tract-precision data to support a tract-level model, across 11 community areas.

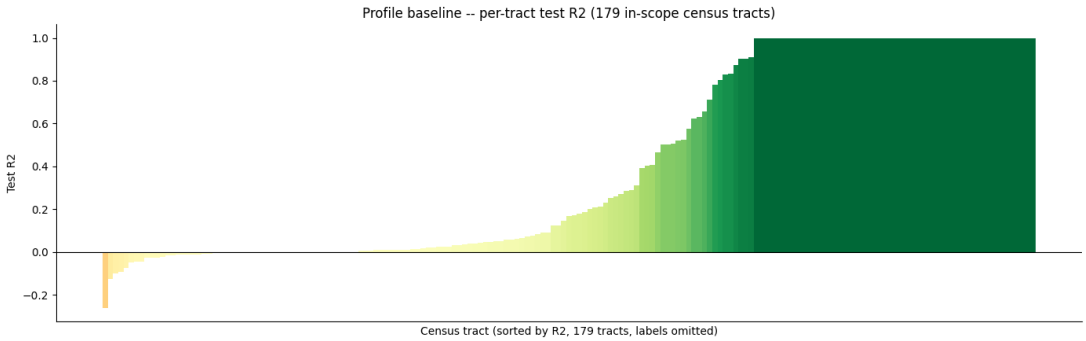


Figure 14: Profile baseline test $R^2$ per census tract. Wider spread than the community-area version (Figure 7), consistent with more units competing for the same total demand.

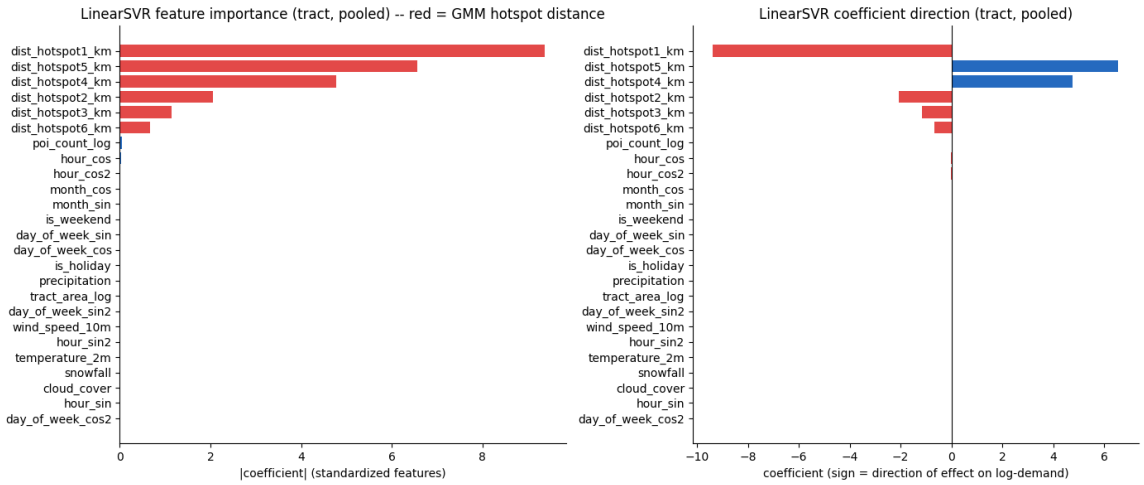


Figure 15: LinearSVR feature-coefficient magnitudes, census tract. GMM hotspot-distance coefficients (red) dominate every other continuous feature.

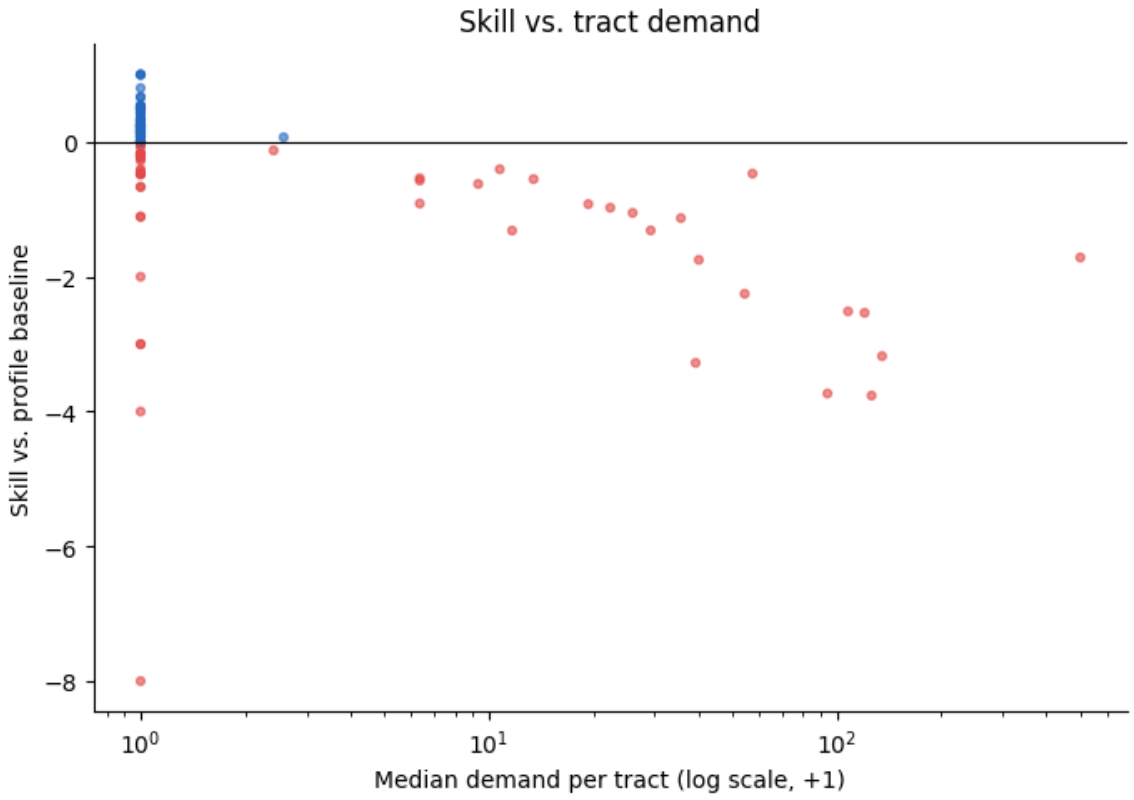


Figure 16: Per-tract skill vs. median demand (log scale).

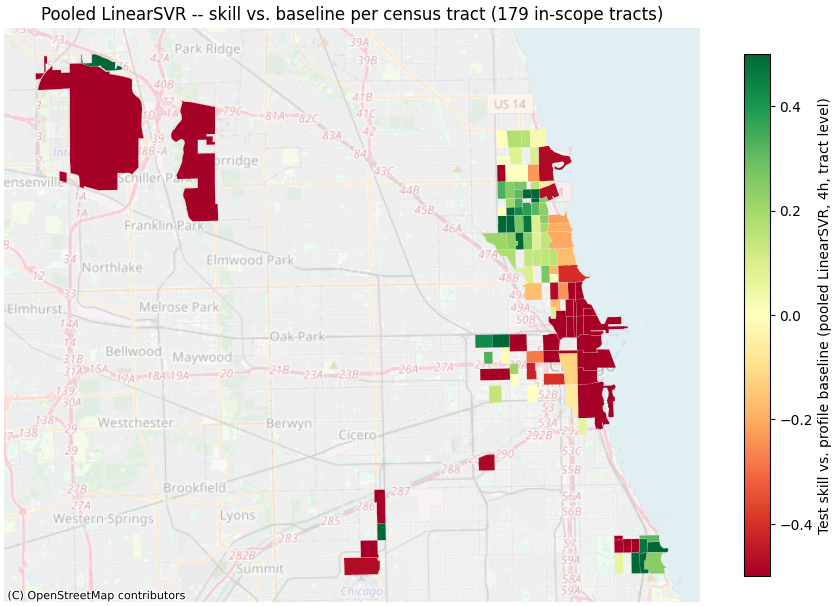


Figure 17: Per-tract skill map, pooled LinearSVR, all 179 in-scope tracts.